Two Papers and Our Project

A beginner's guide to the AVE paper, Look, Listen and Learn, and how AValidate builds on both. No deep-learning background needed.

THE SHORT VERSION

AVE gave the field a dataset and a task: label every second of a video with what is happening, if it can be both seen and heard.
Look, Listen and Learn gave a training trick: show a network a picture and a sound, and ask "do these come from the same video?".
AValidate uses both ideas as building blocks, and turns them into something neither paper made: a tool that tells a person whether the thing on screen is making the sound, and how sure it is, tested more strictly than either paper tested.

PART 0

Ten words you need first

Model	A program that learns a pattern from examples, instead of being given rules.
Training	Showing the model many examples with the right answers until it gets them right on its own.
Label	The right answer attached to an example, like "this second is a dog barking".
Encoder	The part of a model that turns raw input (an image, a sound) into a list of numbers that captures what matters.
Embedding	That list of numbers. Similar things get similar numbers. Think of it as a fingerprint of the content.
Pretrained	A model someone already trained on huge data. We reuse it instead of starting from zero, like hiring someone experienced.
LoRA	A cheap way to adjust a pretrained model. The big model stays frozen; we train small add-on pieces. Like adding sticky notes to a textbook instead of rewriting it.
Head	A small final layer that answers one specific question from the embedding. One model can have several heads.
Positive / negative pair	A positive is a sound and picture that belong together; a negative is one that does not. The model learns from both.
Calibration	Making the model's confidence honest. If it says 80% sure a hundred times, it should be right about eighty times.

PART 1

The AVE paper (Tian et al., ECCV 2018)

What problem were they looking at?

Videos from the internet are messy. A clip labelled “dog barking” may show the dog for only part of the time, and the bark may only happen now and then. The authors wanted models that can say *when* in a video something is both visible and audible. They call that an **audio-visual event**.

What did they build?

A dataset, called AVE. They took about 10,000 YouTube clips from Google’s AudioSet, had trained people watch them, and kept 4,143 ten-second clips across 28 categories: dog bark, guitar, helicopter, man speaking, and so on. For every clip, people marked which seconds contain the event, at one-second resolution. The rest of the seconds are “background”.

AVE: one label for every second of a 10-second clip



A second counts as the event only when the dog is visible AND the bark is audible. Otherwise it is 'BG' (background).

What tasks did they define?

- **Supervised localisation:** for each second, predict the event (one of 28) or “background”. Training uses the per-second labels.
- **Weakly-supervised localisation:** the same, but during training the model only knows the clip’s overall label, not which seconds.
- **Cross-modality localisation:** given a few seconds of audio, find where in the video it fits, or the reverse. For this they trained a network to measure how close an audio segment and a video segment are.

How did their model work, in plain words?

- Each second of video goes through an image network (VGG-19, already trained on ImageNet photos) to get a visual embedding. Each second of audio goes through a sound network already trained on AudioSet.
- An attention step lets the audio point at the part of the image that is probably making the sound.
- A memory-style network (an LSTM) reads the seconds in order, so the model knows what came before.
- A fusion step combines audio and visual, and a final layer predicts the label for each second.

What did they find?

Using audio and video **together** works better than using either alone. Alignment in time matters. Audio can guide where to look in the image.

What does it not do?

- It never answers “is the thing on screen the source of this sound?” as a yes, no or unsure verdict for a user.
- It does not report how trustworthy its confidence is (no calibration).
- Its comparison models are audio-only and video-only models, not “two classifiers that check whether their labels agree”.

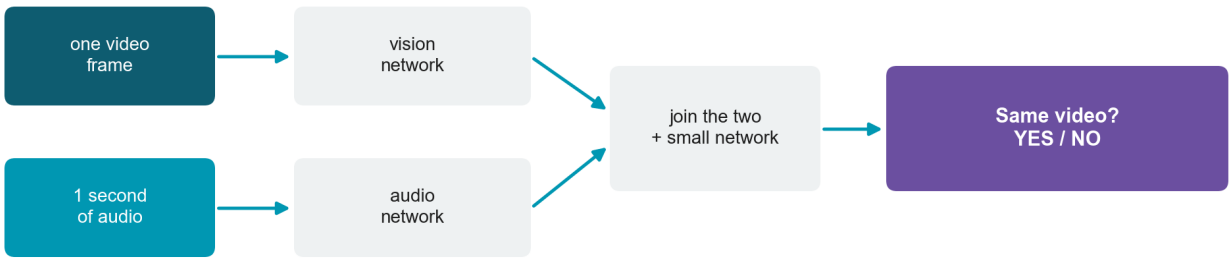
PART 2

Look, Listen and Learn (Arandjelović & Zisserman, ICCV 2017)

What is the idea?

A yes/no game. Show the network one frame from a video and one second of sound. Ask: do these come from the same video? Train it until it gets good at the game.

Look, Listen and Learn: a yes/no game with free labels



YES examples: frame and sound cut from the same moment of the same video.

NO examples: frame from one video, sound from a random other video. Nobody labels anything by hand.

Why is that clever?

The answers come for free. If you cut the frame and the sound from the same moment, the answer is yes. If you take the sound from a random other video, the answer is no. Nobody has to label anything by hand, so you can train on huge amounts of video. Learning from labels you create out of the data itself is called **self-supervised learning**.

What was the real goal?

The game was a means to an end. To win it, the network has to learn what dogs look like and what barking sounds like. The authors cared about those learned **embeddings**, which they then reused for other tasks, like sound classification. The yes/no classifier itself was mostly a training tool.

What does it not do?

- Its “no” examples are random, so most are easy: a dog with traffic noise. It does not deliberately test hard cases like guitar versus banjo, or the right sound at the wrong moment.
- It does not give a calibrated verdict to a user, and has no “uncertain” option.
- It works on one frame and one second, not on second-by-second labels for a whole clip.

ONE MORE IDEA WE BORROW

Later work (Owens & Efros, 2018; Korbar et al., 2018) made the game harder by using the **same** video’s sound, shifted by a few seconds, as a “no” example. Same dog, same bark, wrong moment. That forces the model to pay attention to timing. Our time-shifted negatives come from this.

PART 3

What AValidate does

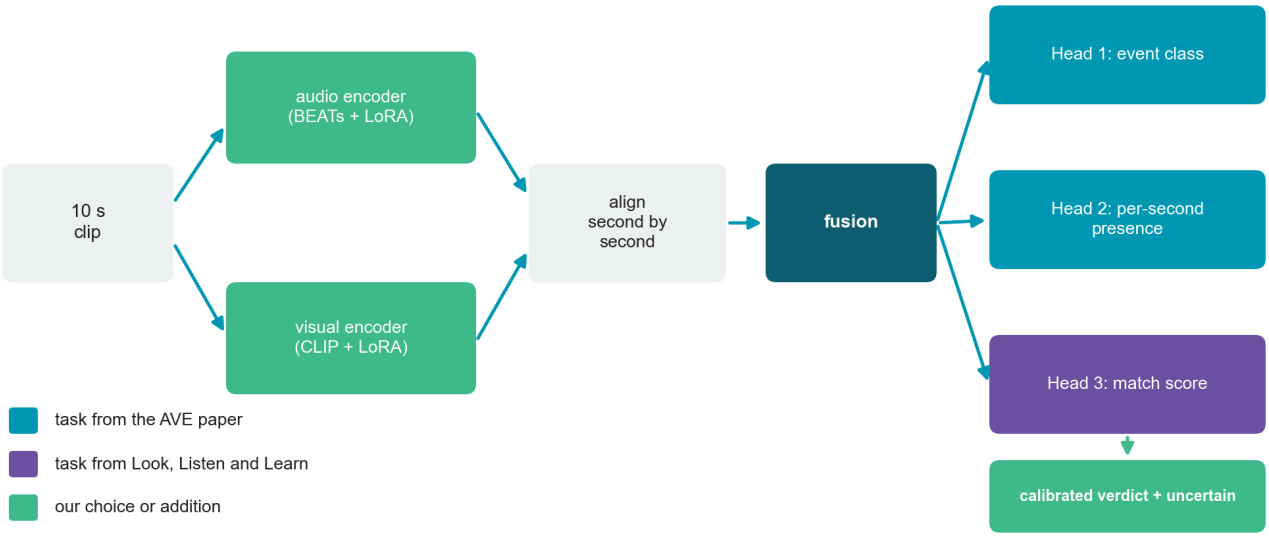
In one sentence?

Given a 10-second clip, AValidate says, second by second, whether the thing on screen is the source of the sound, and gives an honest confidence: strong match, uncertain, or mismatch.

Walk me through one clip.

- A clip comes in: a dog on screen, barking in some seconds.
- The audio encoder turns each second of sound into an embedding; the visual encoder does the same for each second of video. We start these from big pretrained models (BEATs for sound, CLIP for images) and adjust them cheaply with LoRA.
- We line them up: second 3 of audio sits next to second 3 of video.
- Fusion combines each pair into one embedding per second.
- Three heads answer three questions: what is the event (head 1), is it audible and visible in this second (head 2), and do the sound and the picture belong together (head 3).
- The head 3 score is calibrated and turned into a verdict, with “uncertain” for the in-between cases.

AValidate: where each piece comes from



Which pieces come from which paper?

- **Heads 1 and 2** are AVE’s supervised localisation task, split into two questions. We also use AVE’s dataset and its official train/validation/test split.
- **Head 3** is Look, Listen and Learn’s yes/no correspondence game, as is the trick of swapping in audio from other clips to make “no” examples.
- **Time-shifted negatives** come from the synchronisation papers mentioned above.

What is ours

1. A verdict for a person, with honest confidence

AVE predicts labels; Look, Listen and Learn throws its classifier away. We make the yes/no score the product: calibrated so that 80% really means about 80%, and with an “uncertain” band for cases like a person on screen with a dog bark, where a dog may be just out of frame. Neither paper calibrates or allows an unsure answer.

2. A fairer test of whether fusion helps

The obvious shortcut is two separate classifiers: one says “bark”, the other says “dog”, they agree, call it a match. We use that shortcut as our main baseline, and the fused model has to beat it. AVE compared against audio-only and video-only models, which is a different test.

3. Harder exam questions

Random negatives are easy. We score the model separately on easy negatives (dog plus siren), hard ones (guitar video plus banjo sound, both in AVE) and time-shifted ones (right bark, wrong second). That shows whether the model really listens and looks, or just matches categories.

4. Newer building blocks

AVE used 2014-era image features; Look, Listen and Learn trained from scratch. We start from BEATs (2022) and CLIP (2021), adapted with LoRA, which is how modern systems are built on small datasets.

5. One model, one working tool

AVE trains a separate network for each task. We put all three heads on one shared model, and wrap it in a demo where you upload a clip and see the verdict and the per-second timeline.

PART 4

Side by side

	AVE (2018)	Look, Listen and Learn (2017)	AValidate (ours)
Main question	When is the event visible and audible?	Do this frame and sound belong together?	Is the thing on screen the source, and how sure?
Output	A label per second	Learned embeddings (classifier is a tool)	Per-second labels plus a calibrated verdict
Labels	Human, per second	None (made from the data)	AVE labels plus generated pairs
“No” examples	Used only for the matching network	Random other videos	Easy, hard and time-shifted, scored separately
Encoders	Pretrained VGG-era features	Trained from scratch	Pretrained BEATs and CLIP with LoRA
Uncertain answer	No	No	Yes
Calibration	No	No	Yes, measured with ECE
Baseline	Audio-only, video-only	n/a	Two classifiers whose labels agree

PART 5

Are we copying?

Short answer?

No, as long as we cite properly and do not claim the building blocks as ours. Almost every project builds on published methods; what matters is being clear about which parts are borrowed.

An analogy

Think of building a house. The bricks, the cement and the window frames are standard products, and you name the supplier. The design of the house, how it is tested and who it is for, is yours. AVE and Look, Listen and Learn are our suppliers. The verdict, the uncertain band, the stricter tests and the working tool are our design.

When would it become copying?

- Saying “we came up with the idea of swapping audio to make negatives”. That is Look, Listen and Learn’s idea.
- Presenting heads 1 and 2 as a new task. That is AVE’s task.
- Reusing their code or figures without credit.

What we changed to stay on the right side

- The Training Pairs slide now cites Look, Listen and Learn for audio swapping, and [7][8] for time shifts.
- A new “Builds On” slide shows exactly what we take from each paper and what we add.
- Both papers, plus the two synchronisation papers, are in the references.

SAY IT IN 30 SECONDS

“We build on two well-known papers. From AVE we take the dataset and its per-second event task, which become our first two heads. From Look, Listen and Learn we take the idea of learning whether a sound and a picture belong together, which becomes our third head. What we add is a calibrated verdict with an uncertain option, a stricter evaluation against a label-agreement baseline with hard and time-shifted negatives, and newer pretrained encoders combined in one model with a working demo.”

Questions you might get

“Isn’t head 3 just Look, Listen and Learn?”

The task is the same idea, and we cite it. What differs is what we do with it: we calibrate the score, add an uncertain band, train and test on hard and time-shifted negatives, and judge it against a label-agreement baseline.

“Why reuse AVE’s task instead of inventing one?”

Because it lets us compare with published results on the same data and split. Reusing a benchmark is normal and makes results trustworthy.

“What would count as success?”

The fused model beats the two-classifiers-agree baseline, especially on hard and time-shifted negatives, and its confidence is well calibrated.

Sources: Tian et al., ECCV 2018 (arxiv.org/abs/1803.08842; facts checked against the paper on 5 Oct 2026). Arandjelović & Zisserman, ICCV 2017. Owens & Efros, ECCV 2018. Korbar, Tran & Torresani, NeurIPS 2018. BEATs (arxiv.org/abs/2212.09058). CLIP (arxiv.org/abs/2103.00020). LoRA (arxiv.org/abs/2106.09685). The Look, Listen and Learn

and synchronisation-paper details are summarised from well-known published descriptions.